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$$z^4 + (1 - 2i)z^2 - 2i = 0$$

$$\xrightarrow{t=z^2} t^2 + (1 - 2i)t - 2i = 0$$

$$\begin{array}{c|cc} & 1 & (1 - 2i) & -2i \\ -1 & & -1 & 2i \\ \hline & 1 & -2i & 0 \end{array}$$

$$\Leftrightarrow (t + 1)(t - 2i) = 0$$

$$\Rightarrow t_0 = -1 \text{ en } t_1 = 2i$$

$$t_0: z^2 = -1 \Rightarrow z_{1,2} = \pm i$$

$$t_1: z^2 = 2i = (x + yi)^2 = x^2 - y^2 + 2xyi$$

$$\Rightarrow \begin{cases} x^2 - y^2 = 0 \\ 2xy = 2 \end{cases}$$

$$\Rightarrow \begin{cases} x^2 - y^2 = 0 \\ x^2(-y^2) = -1 \end{cases}$$

$$\Rightarrow \lambda^2 - \frac{1}{2} = 0$$

$$\Rightarrow \lambda_{0,1} = \pm 1$$

$$\lambda_0: x^2 = 1 \Rightarrow x = \pm 1$$

$$\lambda_1: -y^2 = -1 \Rightarrow y = \pm 1$$

$$\Rightarrow z_3 = 1 + i$$

$$z_4 = -1 - i$$

References